HYPOTHESIS TESTING

An Intuitive Guide for Making Data Driven Decisions

JIM FROST, MS

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Jim Frost

To contact the author, please email: jim@statisticsbyjim.com.

Visit the author's website at statisticsbyjim.com.

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The best thing about being a statistician is that you get to play in everyone's backyard.

–John Tukey

INTRODUCTION

Goals for this Book

NOTE: This sample contains only the introduction and first two chapters. Please buy the full ebook for all the content listed in the Table of Contents. You can buy it in My Store.

In today's data-driven world, we hear about making decisions based on the data all the time. Hypothesis testing plays a crucial role in that process, whether you're in academia, making business decisions, or in quality improvement. Without hypothesis tests, you risk drawing the wrong conclusions and making bad decisions. That can be costly, either in business dollars or for your reputation as an analyst or scientist.

Chances are high that you'll need a working knowledge of hypothesis testing to produce new findings yourself and to understand the work of others. The world today produces more analyses designed to influence you than ever before. Are you ready for it?

By reading this ebook, you will build a solid foundation for understanding hypothesis tests and become confident that you know when to use each type of test, how to use them properly to obtain reliable results, and how to interpret the results correctly. I present a wide variety of tests that assess characteristics of different data types.

Statistics is the science of learning from data, and hypothesis tests are a vital tool in that process. These tests assess your data using a specific approach to determine whether your results are statistically significant. Hypothesis tests allow you to use a relatively small random sample to draw conclusions about entire populations. That process is fascinating.

I also want you to comprehend what significance truly means in this context. To accomplish these goals, I'm going to teach you how these tests work using an intuitive approach, which helps you fully understand the results.

Hypothesis testing occurs near the end of a long sequence of events. It occurs after you designed your experiment or study, collected a representative sample, randomly assigned subjects, controlled conditions as needed, and collected your data. That sequence of events varies depending on the specifics of your study.

For example, is it a randomized trial or an observational study? Does it involve people? Does your design allow you to identify causation rather than mere correlation? Similarly, the challenges you'll face along the way can vary widely depending on the nature of your subject and the type of data. There are considerations every step of the way that determine whether your study will produce valid results.

Consequently, hypothesis testing builds on a broad range of statistical knowledge, such as inferential statistics, experimental design, measures of central tendency and variability, data types, and probability distributions to name a few. Your hypothesis testing journey will be easier if you are already familiar with these concepts. I'll review some of that information in this book, but I focus on the hypothesis tests. If you need a refresher, consider reading my *Introduction to Statistics* ebook.

You'll notice that there are not many equations in this book. After all, you should let your statistical software handle the calculations while you focus on understanding your results. Consequently, I emphasize the concepts and practices that you'll need to know to perform the analysis and interpret the results correctly. I'll use more graphs than equations! If you need the equations, you'll find them in most textbooks.

In particular, I use many probability distribution plots. Probability distributions are vital components of hypothesis tests. Keep in mind that these plots use complex equations to display the distribution curves and calculate relevant probabilities. I prefer to show you the graphs so you understand the process rather than working through equations!

Throughout this book, I use Minitab statistical software. However, this book is not about teaching particular software but rather how to perform, understand, and interpret hypothesis testing. All common statistical software packages should be able to perform the analyses that I show. There is nothing in here that is unique to Minitab.

CHAPTER 1

Fundamental Concepts

Let's start by cutting to the chase. What is a hypothesis test?

A hypothesis test is a statistical procedure that allows you to use a sample to draw conclusions about an entire population. More specifically, a hypothesis test evaluates two mutually exclusive statements about the population and determines which statement the data support. These two statements are the hypotheses that the procedure tests.

Throughout this book, I'll remind you that these procedures use evidence in samples to make inferences about the characteristics of populations. I want to drive that point home because it's the entire reason for hypothesis testing. Unfortunately, analysts often forget the rationale!

But we're getting ahead of ourselves.

Let's cover some basic hypothesis testing terms that you need to know. We'll cover all these terms in much more detail throughout the book. For now, this chapter provides an overview to show you the relationships between these crucial concepts.

Hypothesis testing is a procedure in inferential statistics. To draw reliable conclusions from a sample, you need to appreciate the differences between descriptive statistics and inferential statistics.

Descriptive vs. Inferential Statistics

Descriptive statistics summarize data for a group that you choose. This process allows you to understand that specific set of observations.

Descriptive statistics describe a sample. That's pretty straightforward. You simply take a group that you're interested in, record data about the group members, and then use summary statistics and graphs to present the group properties. With descriptive statistics, there is no uncertainty because you are describing only the people or items that you actually measure. For instance, if you measure test scores in two classes, you know the precise means for both groups and can state with no uncertainty which one has a higher mean. You're not trying to infer properties about a larger population.

However, if you want to draw inferences about a population, there are suddenly more issues you need to address. We're now moving into inferential statistics. Drawing inferences about a population is particularly important in science where we want to apply the results to a larger population, not just the specific sample in the study. For example, if we're testing a new medication, we don't want to know that it works only for the small, select experimental group. We want to infer that it will be effective for a larger population. We want to generalize the sample results to people outside the sample.

Inferential statistics takes data from a sample and makes inferences about the larger population from which the sample was drawn. Consequently, we need to have confidence that our sample accurately reflects the population. This requirement affects our process. At a broad level, we must do the following:

1. Define the population we are studying.
2. Draw a representative sample from that population.
3. Use analyses that incorporate the sampling error.

We don't get to pick a convenient group. Instead, random sampling allows us to have confidence that the sample represents the population. This process is a primary method for obtaining samples that mirrors the population on average. Random sampling produces statistics, such as the mean, that do not tend to be too high or too low. Using a random sample, we can generalize from the sample to the broader population.

While samples are much more practical and less expensive to work with, there are tradeoffs. Typically, we learn about the population by drawing a relatively small sample from it. We are a very long way off from measuring all people or objects in that population. Consequently, when you estimate the properties of a population from a sample, the sample statistics are unlikely to equal the actual population value exactly. For instance, your sample mean is unlikely to equal the population mean. The difference between the sample statistic and the population value is the sampling error.

You gain tremendous benefits by working with a random sample drawn from a population. In most cases, it is simply impossible to measure the entire population to understand its properties. The alternative is to gather a random sample and then use hypothesis testing to analyze the sample data. However, a crucial point to remember is that hypothesis tests make assumptions about the data collection process. For instance, these tests assume that the data were collected using a method that tends to produce representative samples. After all, if the sample isn't similar to the population, you won't be able to use the sample to draw conclusions about the population.

Random sampling is the most commonly known method for obtaining an unbiased, representative sample, but there are other techniques.

That discussion goes beyond this book, but my *Introduction to Statistics* book describes some of the other procedures.

Population Parameters vs. Sample Statistics

A parameter is a value that describes a characteristic of an entire population, such as the population mean. Because you can rarely measure an entire population, you usually don't know the real value of a parameter. In fact, parameter values are almost always unknowable. While we don't know the value, it definitely exists.

For example, the average height of adult women in the United States is a parameter that has an exact value—we just don't know what it is!

The population mean and standard deviation are two common parameters. In statistics, Greek symbols usually represent population parameters, such as μ (mu) for the mean and σ (sigma) for the standard deviation.

A statistic is a characteristic of a sample. If you collect a sample and calculate the mean and standard deviation, these are sample statistics. Inferential statistics allow you to use sample statistics to make conclusions about a population. However, to draw valid conclusions, you must use representative sampling techniques. These techniques help ensure that samples produce unbiased estimates. Biased estimates are systematically too high or too low. You want unbiased estimates because they are correct on average. Use random sampling and other representative sampling methodologies to obtain unbiased estimates.

In inferential statistics, we use sample statistics to estimate population parameters. For example, if we collect a random sample of adult women in the United States and measure their heights, we can calculate the sample mean and use it as an unbiased estimate of the population mean. We can also create confidence intervals to obtain a range that the actual population value likely falls within.

Population Parameter	Sample Statistic
Mu (μ)	Sample mean
Sigma (σ)	Sample standard deviation

Random Sampling Error

When you have a representative sample, the sample mean and other characteristics are unlikely to equal the population values exactly. The sample is similar to the population, but it is never identical to the population.

The differences between sample statistics and population parameters are known as sampling error. If you want to use samples to make inferences about populations, you need statistical methods that incorporate estimates of the sampling error. As you'll learn, sampling error blurs the line between real effects and random variations caused by sampling. Hypothesis testing helps you separate those two possibilities.

Because population parameters are unknown, we also never know sampling error exactly. However, using hypothesis testing, we can estimate the error and factor it into the test results.

Parametric versus Nonparametric Analyses

Parametric statistics is a branch of statistics that assumes sample data come from populations that are adequately modeled by probability distributions with a set of parameters. Parametric analyses are the most common statistical methods and this book focuses on them. Consequently, you will see many references to probability distributions, probability distribution plots, parameter estimates, and assumptions about your data following a particular distribution (often the normal distribution) throughout this book.

Conversely, nonparametric tests don't assume that your data follow a particular distribution. While this book doesn't emphasize those methods, I cover some of them in the last chapter so you can see how they compare and have an idea about when to use them. Statisticians use nonparametric analyses much less frequently than their parametric counterparts.

Hypothesis Testing

Hypothesis testing is a statistical analysis that uses sample data to assess two mutually exclusive theories about the properties of a population. Statisticians call these theories the null hypothesis and the alternative hypothesis. A hypothesis test assesses your sample statistic and factors in an estimate of the sampling error to determine which hypothesis the data support.

When you can reject the null hypothesis, the results are statistically significant, and your data support the theory that an effect exists at the population level.

Hypothesis tests use sample data answer questions like the following:

- Is the population mean greater than or less than a particular value?
- Are the means of two or more populations different from each other?

For example, if we study the effectiveness of a new medication by comparing the outcomes in a treatment and control group, hypothesis tests can tell us whether the drug's effect that we observe in the sample is likely to exist in the population. After all, we don't want to use the medication if it is effective only in our specific sample. Instead, we need evidence that it'll be useful in the entire population of patients. Hypothesis tests allow us to draw these types of conclusions about whole populations.

Null Hypothesis

In hypothesis testing, the null hypothesis is one of two mutually exclusive theories about the population's properties. Typically, the null hypothesis states there is no effect (i.e., the effect size equals zero). H_0 often signifies the null.

In all hypothesis testing, the researchers are testing an effect of some sort. Effects can be the effectiveness of a new vaccination, the durability of a new product, the proportion of defects in a manufacturing process, and so on. There is some benefit or difference that the researchers hope to identify.

However, there might be no effect or no difference between the experimental groups. In statistics, we call this lack of an effect the null hypothesis. Therefore, if you can reject the null, you can favor the alternative hypothesis, which states that the effect exists (doesn't equal zero) at the population level.

You can think of the null as the default theory that requires sufficiently strong evidence in your sample to be able to reject it.

For example, when you're comparing the means of two groups, the null often states that the difference between the two means equals zero. In other words, the groups are not different.

Alternative Hypothesis

The alternative hypothesis is the other theory about the properties of the population in hypothesis testing. Typically, the alternative hypothesis states that a population parameter does not equal the null hypothesis value. In other words, there is a non-zero effect. If your sample contains sufficient evidence, you can reject the null and favor the alternative hypothesis. H_1 or H_A usually identifies the alternative.

For example, if you're comparing the means of two groups, the alternative hypothesis often states that the difference between the two means does not equal zero.

Effect

The effect is the difference between the population value and the null hypothesis value. The effect is also known as population effect or the difference. For example, the mean difference between the health outcome for a treatment group and a control group is the effect.

Typically, you do not know the size of the actual effect. However, you can use a hypothesis test to determine whether an effect exists and estimate its size.

For example, if the mean of one group is 10 and the mean of another group is 2, the effect is 8.

Significance Level (Alpha)

The significance level defines how strong the sample evidence must be to conclude an effect exists in the population.

The significance level, also known as alpha or α, is an evidentiary standard that researchers set before the study. It specifies how strongly the sample evidence must contradict the null hypothesis before you can reject the null for the entire population. This standard is defined by the probability of rejecting a true null hypothesis. In other words, it is the probability that you say there is an effect when there is no effect. Lower significance levels indicate that you require more substantial evidence before you will reject the null.

For instance, a significance level of 0.05 signifies a 5% risk of deciding that an effect exists when it does not exist.

Use p-values and significance levels together to determine which hypothesis the data support, as described in the p-value section.

P-values

P-values indicate the strength of the sample evidence against the null hypothesis. If it is less than the significance level, your results are statistically significant.

P-values are the probability that you would obtain the effect observed in your sample, or larger, if the null hypothesis is correct. In simpler terms, p-values tell you how strongly your sample data contradict the null. Lower p-values represent stronger evidence against the null.

If the p-value is less than or equal to the significance level, you reject the null hypothesis and your results are statistically significant. The data support the alternative hypothesis that the effect exists in the population. When the p-value is greater than the significance level, your sample data don't provide enough evidence to conclude that the effect exists.

Here's the statistical terminology for these decisions.

- When the p-value is less than or equal to the significance level, you reject the null hypothesis.
- When the p-value is greater than the significance level, you fail to reject the null hypothesis.

If you need help remembering this general rule about comparing p-values to significance levels, here are two mnemonic phrases:

- When the p-value is low, the null must go.
- If the p-value is high, the null will fly.

Statistical Significance

When your p-value is less than the significance level, your results are statistically significant. This condition indicates the strength of the evidence in your sample (p-value) exceeds the evidentiary standard you defined (significance level). Your sample evidence provides sufficient evidence to conclude that the effect exists in the population.

Confidence intervals (CIs)

In inferential statistics, a principal goal is to estimate population parameters. These parameters are the unknown values for the entire population, such as the population mean and standard deviation. These parameter values are not only unknown but almost always unknowable. Typically, it's impossible to measure an entire population. The sampling error I mentioned earlier produces uncertainty, or a margin of error, around our estimates.

Suppose we define our population as all high school basketball players. Then, we draw a random sample from this population and calculate the mean height of 181 cm. This sample estimate of 181 cm is the best estimate of the mean height of the population. Because the mean is from a sample, it's virtually guaranteed that our estimate of the population parameter is not exactly correct.

Confidence intervals incorporate the uncertainty and sample error to create a range of values the actual population value is like to fall within. For example, a confidence interval of [176 186] indicates that we can be confident that the real population mean falls within this range.

Significance Levels In-Depth

Before getting to the first example of a hypothesis test, I want you to understand significance levels conceptually. It lies at the heart of how we use statistics to learn. How do we determine that we have significant results?

Significance levels in statistics are a crucial component of hypothesis testing. However, unlike other values in your statistical output, the significance level is not something that statistical software calculates. Instead, you choose the significance level. Why is that?

In this section, I'll explain the significance level, why you choose its value, and how to choose a good value.

Your sample data provide evidence for an effect. The significance level is a measure of how strong the sample evidence must be before determining the results are statistically significant. It defines the line between the evidence being strong enough to conclude that the effect exists in the population versus it's weak enough that we can't rule out the possibility that the sample effect is just random sampling error. Because we're talking about evidence, let's look at a courtroom analogy.

Evidentiary Standards in the Courtroom

Criminal cases and civil cases vary greatly, but both require a minimum amount of evidence to convince a judge or jury to prove a claim against the defendant. Prosecutors in criminal cases must prove the defendant is guilty "beyond a reasonable doubt," whereas plaintiffs in a civil case must present a "preponderance of the evidence." These terms are evidentiary standards that reflect the amount of evidence that civil and criminal cases require.

For civil cases, most scholars define a preponderance of evidence as meaning that at least 51% of the evidence shown supports the plaintiff's claim. However, criminal cases are more severe and require more substantial evidence, which must go beyond a reasonable doubt. Most scholars define that evidentiary standard as being 90%, 95%, or even 99% sure that the defendant is guilty.

In statistics, the significance level is the evidentiary standard. For researchers to successfully make the case that the effect exists in the population, the sample must contain sufficient evidence.

In court cases, you have evidentiary standards because you don't want to convict innocent people.

In hypothesis tests, we have the significance level because we don't want to claim that an effect or relationship exists when it does not exist.

Significance Levels as an Evidentiary Standard

In statistics, the significance level defines the strength of evidence in probabilistic terms. Specifically, alpha represents the probability that tests will produce statistically significant results when the null hypothesis is correct. You can think of this error rate as the probability of a false positive. The test results lead you to believe that an effect exists when it actually does not exist.

Obviously, when the null hypothesis is correct, we want a low probability that hypothesis tests will produce statistically significant results. For example, if alpha is 0.05, your analysis has a 5% chance of a significant outcome when the null hypothesis is correct.

Just as the evidentiary standard varies by the type of court case, you can set the significance level for a hypothesis test depending on the consequences of a false positive. By changing alpha, you increase or decrease the amount of evidence you require in the sample to conclude that the effect exists in the population.

Changing Significance Levels

Because 0.05 is the standard alpha, we'll start by adjusting away from that value. Typically, you'll need a good reason to change the significance level to something other than 0.05. Also, note the inverse relationship between alpha and the amount of required evidence. For instance, increasing the significance level from 0.05 to 0.10 lowers the evidentiary standard. Conversely, decreasing it from 0.05 to 0.01 increases the bar. Let's look at why you would consider changing alpha and how it affects your hypothesis test.

Increasing the Significance Level

Imagine you're testing the strength of party balloons. You'll use the test results to determine which brand of balloons to buy. A false positive here leads you to buy balloons that are not stronger. The drawbacks of a false positive are very low. Consequently, you could consider lessening the amount of evidence required by changing the significance level to 0.10. Because this change decreases the amount of evidence needed, it makes your test more sensitive to detecting differences, but it also increases the chance of a false positive from 5% to 10%.

Decreasing the Significance Level

Conversely, imagine you're testing the strength of fabric for hot air balloons. A false positive here is very risky because lives are on the line! You want to be very confident that the material from one manufacturer is stronger than the other. In this case, you should increase the amount of evidence required by changing alpha to 0.01. Because this change increases the amount of evidence needed, it makes your test less sensitive to detecting differences, but it decreases the chance of a false positive from 5% to 1%.

It's all about the tradeoff between sensitivity and false positives!

In conclusion, a significance level of 0.05 is the most common. However, it's the analyst's responsibility to determine how much evidence to require for concluding that an effect exists. How problematic is a false positive? There is no single correct answer for all circumstances. Consequently, you need to choose the significance level!

While significance levels indicate the amount of evidence required, p-values represent the strength of the evidence in your sample. When your p-value is less than or equal to the significance level, the strength of the sample evidence meets or exceeds your evidentiary standard for rejecting the null hypothesis and concluding that the effect exists.

How Hypothesis Tests Work

Here's the approach I'll use throughout the book. I'll start with this example that compares a sample mean to a target value. It's an excellent place to start because it's a relatively simple test. I'll intentionally gloss over some of the finer details for now so you can focus on the basics of why you use this test, how it works in a general sense, and how to interpret the results.

In later chapters, I'll add in more of the details and considerations that we're skipping here. At first, I'll stick mainly with hypothesis tests that compare group means because they are the most common. After covering the basics of tests for means, you'll understand the fundamental mechanics for all hypothesis tests. At that point, I'll switch to other types of hypothesis tests and help you determine which ones to use for your data.

Hypothesis testing is a vital process in inferential statistics where the goal is to use sample data to draw conclusions about an entire population. In the testing process, you use significance levels and p-values to determine whether the test results are statistically significant. In this section, I'll keep the discussion more general to avoid using terminology that we haven't covered yet, although you will learn about one new concept—sampling distributions.

You hear about results being statistically significant all the time. But, what do significance levels, p-values, and statistical significance actually represent? Why do we even need to use hypothesis tests in statistics?

I'll answer all of these questions using graphs and concepts to explain how hypothesis tests function to provide a more intuitive explanation.

Let's start by understanding why we need to use hypothesis tests.

A researcher is studying fuel expenditures for families and wants to determine if the monthly cost had changed since last year when the average was $260 per month. The researcher draws a random sample of 25 families and analyzes their monthly costs for this year. You can download the CSV data file: FuelCosts. Below are the descriptive statistics for this year.

Descriptive Statistics: Fuel Cost

Variable	Total Count	Mean	SE Mean	StDev
Fuel Cost	25	330.6	30.8	154.2

We'll build on this example to answer the research question and show how hypothesis tests work.

Descriptive Statistics Won't Answer the Question

The researcher collected a random sample and found that this year's sample mean (330.6) is greater than last year's mean (260). Why perform a hypothesis test at all? We can see that this year's mean is higher by $70! Isn't that different?

Regrettably, the situation isn't as straightforward as you might think because we're analyzing a sample instead of the full population. There are huge benefits when working with samples because it is usually impossible to collect data from an entire population. However, the tradeoff for working with a manageable sample requires that we account for sample error.

The sampling error is the gap between the sample statistic and the population parameter. For our example, the sample statistic is the sample mean, which is 330.6. The population parameter is μ, or mu, which is the average of the entire population. Unfortunately, the value of the population parameter is not only unknown but usually unknowable.

We obtained a sample mean of 330.6. However, it's conceivable that, due to sampling error, the mean of the population might be only 260. If the researcher drew another random sample, the next sample mean might be closer to 260. It's impossible to assess this possibility by looking at only the sample mean.

We need to use a hypothesis test to determine the likelihood of obtaining our sample mean if the population mean is 260.

A Sampling Distribution Determines Whether Our Sample Mean is Unlikely

It is improbable for any sample mean to equal the population mean because of sample error. In our case, the sample mean of 330.6 is almost definitely not equal to the population mean for fuel expenditures.

If we could obtain a substantial number of random samples and calculate the sample mean for each sample, we'd observe a broad spectrum of sample means. We'd even be able to graph the distribution of sample means from this process.

This type of distribution is called a sampling distribution. You obtain a sampling distribution by drawing many random samples of the same size from the same population. Why the heck would we do this?

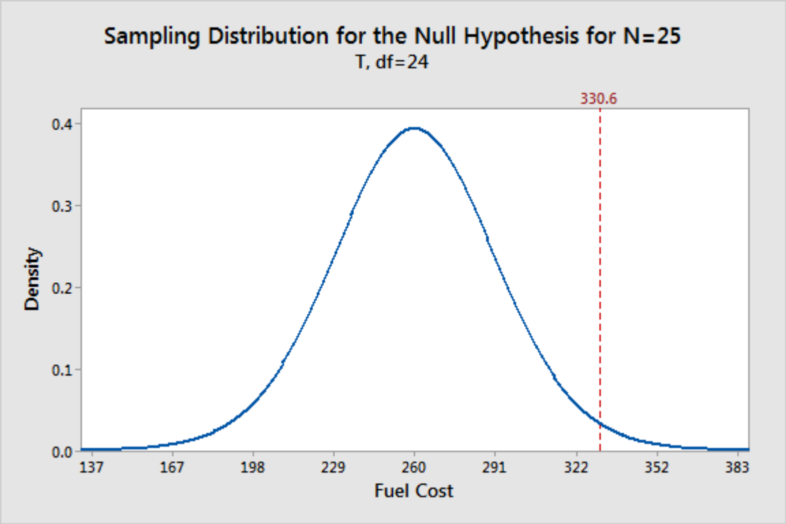
Because sampling distributions allow you to determine the likelihood of obtaining your sample statistic and they're crucial for performing hypothesis tests.

Luckily, we don't need to go to the trouble of collecting numerous random samples! Statistical procedures estimate sampling distributions using the properties of samples. In chapter 3, I'll show this process in-depth. For now, I want you to focus on the idea that the one sample the study collected is only one of an infinite number of potential samples it could have drawn. That's a crucial concept in hypothesis testing and inferential statistics.

We want to find out if the average fuel expenditure this year (330.6) is different from last year (260). To answer this question, we'll graph the sampling distribution based on the assumption that the mean fuel cost for the entire population has not changed and is still 260. Hypothesis tests always use sampling distributions that assume the null hypothesis is correct. Likewise, we use the null hypothesis value as the basis of comparison for our observed sample value.

Graphing our Sample Mean in the Context of the Sampling Distribution

The graph below shows which sample means are more likely and less likely if the population mean is 260. We can place our sample mean in this distribution. This broader context helps us see how unlikely our sample mean is if the null hypothesis is correct (μ = 260).



The graph displays the estimated distribution of sample means. The most likely values are near 260 because the plot assumes that this is the real population mean. However, given random sampling error, it would not be surprising to observe sample means ranging from 167 to 352. If the population mean is still 260, our observed sample mean (330.6) isn't the most likely value, but it's not entirely implausible either.

Sampling distributions are a type of probability distribution plot, which I discuss in more detail in my *Introduction to Statistics* book. On a probability plot, the entire area under the distribution curve equals 1. The proportion of the area under a curve that corresponds to a range of values along the X-axis represents the likelihood a value will fall within that range. Hypothesis tests use the ability of probability distributions to calculate probabilities for ranges of values to determine statistical significance.

The sampling distribution indicates that we are relatively unlikely to obtain a sample of 330.6 if the population mean is 260. Is our sample mean *so* unlikely that we can reject the notion that the population mean is 260?

In statistics, we call this rejecting the null hypothesis. If we reject the null for our example, the difference between the sample mean (330.6) and 260 is statistically significant. In other words, the sample data favor the hypothesis that the population average does *not* equal 260.

However, look at the sampling distribution chart again. Notice that there is no particular location on the curve where you can definitively draw this conclusion. There is only a consistent decrease in the likelihood of observing sample means that are farther from the null hypothesis value. Where do we decide a sample mean is far away enough?

To answer this question, we'll need more hypothesis testing tools! The hypothesis testing procedure quantifies our sample's unusualness with a probability and then compares it to an evidentiary standard. This process allows you to make an objective decision about the strength of the evidence.

We're going to add the tools we need to make this decision to the graph—significance levels and p-values!

These tools allow us to test these two hypotheses:

- Null hypothesis: The population mean equals the null hypothesis mean (260).
- Alternative hypothesis: The population mean does not equal the null hypothesis mean (260).

Graphing Significance Levels as Critical Regions

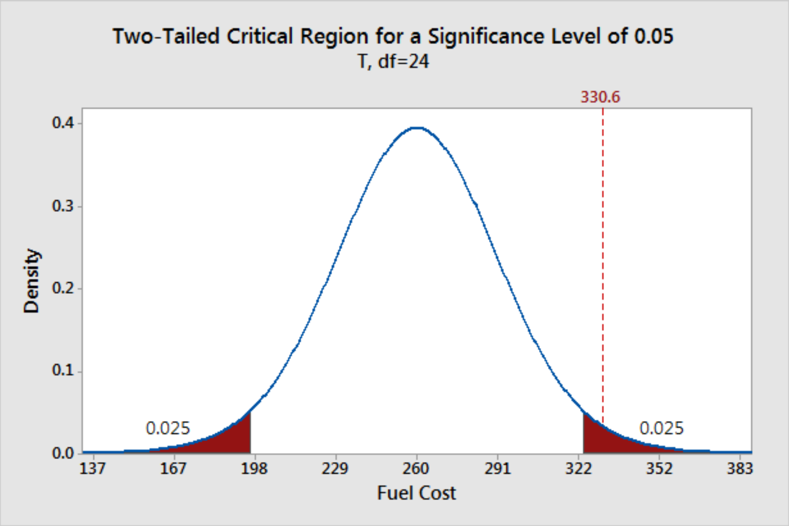
A significance level, also known as alpha or α, is an evidentiary standard that a researcher sets before the study. It defines how strongly the sample evidence must contradict the null hypothesis before you can reject the null hypothesis for the entire population. The strength of the evidence is determined by the probability of rejecting a true null hypothesis.

Lower significance levels require stronger sample evidence to be able to reject the null hypothesis. For example, to be statistically significant at the 0.01 significance level requires more substantial evidence than the 0.05 significance level.

The technical nature of these types of questions can make your head spin. A picture can bring these ideas to life!

On the probability distribution plot, the significance level defines how far the sample value must be from the null value before we can reject the null hypothesis. The percentage of the area under the curve that is shaded equals the probability that the sample value will fall in those regions if the null hypothesis is correct.

To represent a significance level of 0.05, I'll shade 5% of the distribution furthest from the null value.



The two shaded regions in the graph are equidistant from the central value of the null hypothesis. Each region has a probability of 0.025, which sums to our desired total of 0.05. These shaded areas are called the critical regions for a two-tailed hypothesis test.

The critical region defines sample values that are improbable enough to warrant rejecting the null hypothesis. If the null hypothesis is correct and the population mean is 260, random samples (n=25) from this population have means that fall in the critical regions 5% of the time.

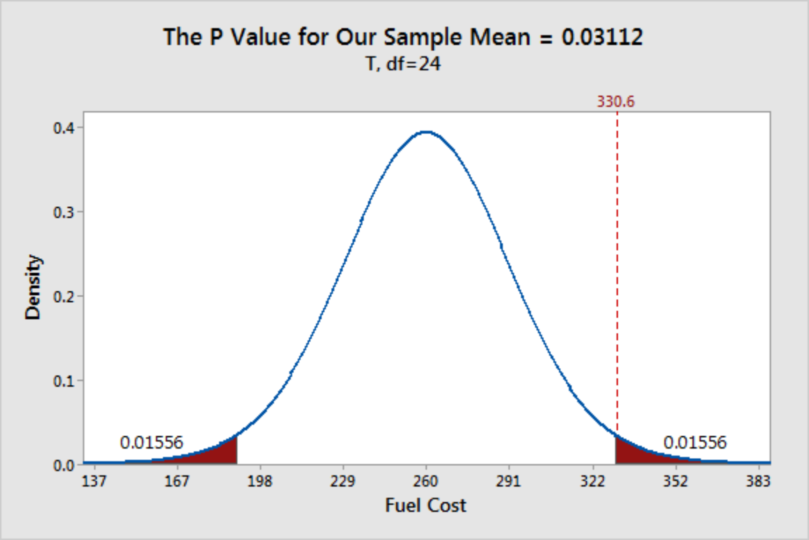
Our sample mean is statistically significant at the 0.05 level because it falls in the critical region.

What Are P-values?

P-values are the probability that a sample will have an effect at least as extreme as the effect observed in your sample *if* the null hypothesis is correct.

This tortuous, technical definition for p-values can make your head spin. Let's graph it!

First, we need to calculate the effect that is present in our sample. The effect is the distance between the sample value and the null value: 330.6 – 260 = 70.6. Next, I'll shade the regions on both sides of the distribution that are at least as far away as 70.6 from the null (260 +/- 70.6). This process graphs the probability of observing a sample mean at least as extreme as our sample mean.



The total probability of the two shaded regions is 0.03112. If the null hypothesis value (260) is true and you drew many random samples, you'd expect sample means to fall in the shaded regions about 3.1% of the time. In other words, you will observe sample effects at least as large as 70.6 about 3.1% of the time if the null hypothesis is correct. That's the p-value!

There's a reason why I'm shading both tails. We'll come back to this in chapter 6 when compare one-tailed and two-tailed tests!

Using P-values and Significance Levels Together

If your p-value is less than or equal to your alpha level, reject the null hypothesis.

The p-value results are consistent with our graphical representation. The p-value of 0.03112 is significant at the alpha level of 0.05.

With a significance level of 0.05, the sample effect is statistically significant. Our data support the alternative hypothesis, which states that the population mean doesn't equal 260. We can conclude that mean fuel expenditures have increased since last year.

Chapter 4 discusses the proper interpretation of p-values in much more detail.

Discussion about Statistically Significant Results

Hypothesis tests determine whether your sample data provide evidence that is strong enough to reject the null hypothesis for the entire population. The test compares your sample statistic to the null value and determines whether it is sufficiently rare. "Sufficiently rare" is defined in a hypothesis test by:

- Assuming that the null hypothesis is correct—the graphs center on the null value.
- The significance (alpha) level—how far out from the null value is the critical region?
- The sample statistic—is it within the critical region?

There is no special significance level that correctly determines which studies have real population effects 100% of the time. The traditional significance levels of 0.05 and 0.01 are attempts to manage the tradeoff between having a low probability of rejecting a true null hypothesis and detecting an effect if one actually exists.

The significance level is the rate at which you incorrectly reject null hypotheses that are actually true. For example, for all studies that use a significance level of 0.05 and the null hypothesis is correct, you can expect 5% of them to have sample statistics that fall in the critical region. When this error occurs, you aren't aware that the null hypothesis is correct, but you'll reject it because the p-value is less than 0.05.

This error does not indicate that the researcher made a mistake. As the graphs show, you can observe extreme sample statistics due to sample error alone. It's the luck of the draw!

Hypothesis tests are crucial when you want to use sample data to make conclusions about a population because these tests account for sample error. Using significance levels and p-values to determine when to reject the null hypothesis improves the probability that you will draw the correct conclusion.

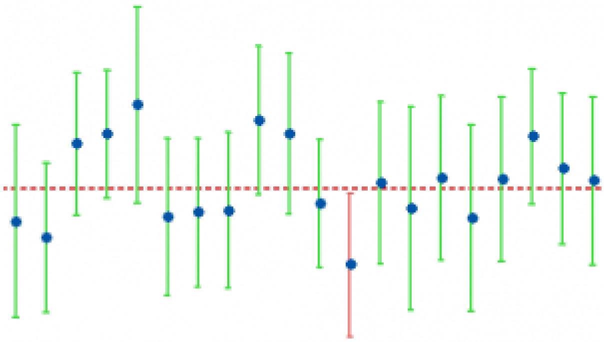
How Confidence Intervals Work

A confidence interval is calculated from a sample and provides a range of values that likely contains the unknown value of a population parameter. In this section, I demonstrate how confidence intervals and confidence levels work using graphs and concepts. In the process, you'll see how confidence intervals are very similar to p-values and significance levels.

You can calculate a confidence interval from a sample to obtain a range for where the population parameter is likely to reside. For example, a confidence interval of [9 11] indicates that the population mean is likely to be between 9 and 11.

Different random samples drawn from the same population are liable to produce slightly different intervals. If you draw many random samples and calculate a confidence interval for each sample, a specific proportion of the intervals contain the population parameter. That percentage is the confidence level.

For example, a 95% confidence level suggests that if you draw 20 random samples from the same population, you'd expect 19 of the confidence intervals to include the population value, as shown below.



The confidence interval procedure provides meaningful estimates because it produces ranges that usually contain the parameter.

We'll create a confidence interval for the population mean using the fuel cost example that we've been developing. With other types of data, you can create intervals for proportions, frequencies, regression coefficients, and differences between populations.

Precision of the Estimate

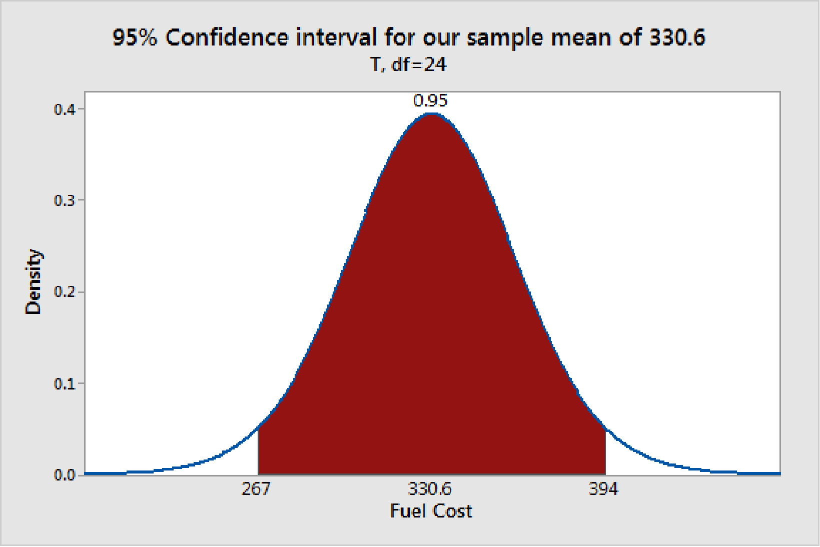
Confidence intervals include the point estimate for the sample with a margin of error around the point estimate. The point estimate is the most likely value of the parameter and equals the sample value. The margin of error accounts for the amount of doubt involved in estimating the population parameter. The more variability there is in the sample data, the less precise the estimate, which causes the margin of error to extend further out from the point estimate. Confidence intervals help you navigate the uncertainty of how well a sample estimates a value for an entire population.

With this in mind, confidence intervals can help you compare the precision of different estimates. Suppose two studies estimate the same mean of 10. It appears like they obtained the same results. However, using 95% confidence intervals, we see that one interval is [5 15] while the other is [9 11]. The latter confidence interval is narrower, which suggests that it is a more precise estimate.

Graphical Representation

Let's delve into how confidence intervals incorporate the margin of error. Like the previous sections, I'll use the same sampling distribution that showed us how hypothesis tests work.

There are two critical differences between the sampling distribution graphs for significance levels and confidence intervals. The significance level chart centers on the null value, and we shade the outside 5% of the distribution. Conversely, the confidence interval graph centers on the sample mean, and we shade the center 95% of the distribution.



The shaded range of sample means [267 392] covers 95% of this sampling distribution. This range is the 95% confidence interval for our sample data. We can be 95% confident that the population mean for fuel costs fall between 267 and 392.

The graph emphasizes the role of uncertainty around the point estimate. This graph centers on our sample mean. If the population mean equals our sample mean, random samples (N=25) from this population will fall within this range 95% of the time.

We don't really know whether our sample mean is near the population mean. However, we know that the sample mean is an unbiased estimate of the population mean. An unbiased estimate is one that doesn't tend to be too high or too low. It's correct on average. Confidence intervals are correct on average because they use sample estimates that are correct on average. Given what we know, the sample mean is the most likely value for the population mean.

Given the sampling distribution, it would not be unusual for other random samples drawn from the same population to have means that fall within the shaded area. In other words, given that we did obtain the sample mean of 330.6, it would not be surprising to get other sample means within the shaded range.

If these other sample means would not be unusual, then we must conclude that these other values are also likely candidates for the population mean. There is inherent uncertainty when you use sample data to make inferences about an entire population. Confidence intervals help you gauge the amount of uncertainty in your sample estimates.

Confidence Intervals and P-values Always Agree

If you want to determine whether your test results are statistically significant, you can use either p-values with significance levels or confidence intervals. These two approaches always agree.

The relationship between the confidence level and the significance level for a hypothesis test is as follows:

Confidence level = 1 – Significance level (alpha)

For example, if your significance level is 0.05, the equivalent confidence level is 95%.

Both of the following conditions represent a hypothesis test with statistically significant results:

- The p-value is smaller than the significance level.
- The confidence interval excludes the null hypothesis value.

Further, it is always true that when the p-value is less than your significance level, the interval excludes the value of the null hypothesis.

In the fuel cost example, our hypothesis test results are statistically significant because the p-value (0.03112) is less than the significance level (0.05). Likewise, the 95% confidence interval [267 394] excludes the null hypothesis value (260). Using either method, we draw the same conclusion.

The p-value and confidence interval results always agree. To understand the basis of this agreement, we need to remember how confidence levels and significance levels function:

- A confidence level determines the distance between the sample mean and the confidence limits.
- A significance level determines the distance between the sample mean and the critical regions.

Both concepts specify a distance from the mean to a limit. Surprise! These distances are precisely the same length.

The hypothesis test in this example is a 1-sample t-test, which calculates this distance as follows:

The critical t-value * standard error of the mean

Interpreting these statistics goes beyond the scope of this section. But, using this equation, the distance for our fuel cost example is $63.57.

P-value and significance level approach: If the sample mean is more than $63.57 from the null hypothesis mean, the sample mean falls within the critical region, and the difference is statistically significant.

Confidence interval approach: If the null hypothesis mean is more than $63.57 from the sample mean, the interval does not contain this value, and the difference is statistically significant.

Of course, they always agree!

When the same hypothesis test generates the p-values and confidence intervals, and you use an equivalent confidence level and significance level, the two approaches always agree.

I Really Like Confidence Intervals!

Analysts often place more emphasis on using p-values to determine whether a result is statistically significant. Unfortunately, a statistically significant effect might not always be practically meaningful. For example, a significant effect can be too small to be of any importance in the real world.

You should always consider both the size and precision of the estimated effect. Ideally, an estimated effect is both large enough to be meaningful and sufficiently precise for you to trust. Confidence intervals allow you to assess both of these considerations!

Review and Next Steps

This chapter covered the essential differences between descriptive and inferential statistics. If you want to generalize your results beyond a sample to a larger population, you'll need to use a representative sampling methodology and a test that factors in sampling error. You don't want to confuse sampling error with a real effect! Hypothesis tests help you make the correct decisions.

After covering some basic terminology and concepts, I presented a simplified version of a hypothesis test. In this example, I introduced the concept of a sampling distribution. Hypothesis testing treats the sample that a study collects as only one of an infinite number that the study could have collected. By constructing a sampling distribution that assumes the null hypothesis is correct, we can determine how unusual our data are if the null is true.

While the hypothesis test in this chapter was unnamed, it is a 1-sample t-test because it compared a single sample mean to a target value. Chapter 2 explores t-tests in more detail. You'll learn about the different types of t-tests and when to use each one, the assumptions and conditions that your data must satisfy to produce reliable results, and how to interpret these t-tests.

CHAPTER 2

T-Test Uses, Assumptions, and Analyses

The example in the previous chapter used a 1-sample t-test. In this chapter, we'll explore t-tests in greater detail. I want you to get your feet wet quickly and know how to interpret the statistical output for t-tests. Then, we'll dive into a deeper understanding of how t-tests work behind the scenes.

T-tests are hypothesis tests that assess the means of one or two groups. Depending on the t-test and how you configure it, the test can determine whether:

- One mean is different from a hypothesized value.
- Two group means are different.
- Paired means are different.

In this chapter, you will learn about three t-tests, 1-sample t-tests, 2-sample tests, and paired t-tests. When statisticians talk about one-sample and two-sample tests, we're referring to how many groups we're comparing. One-sample tests have one group and the test compares the group mean to a target value. That was the type of test in the energy cost example. Two-sample tests compare two groups. Are the means equal or different?

t-Tests can compare the means for up to two groups. If you have three or more groups, you'll need to use ANOVA. That's a topic for a later chapter!

Before we get to the t-tests themselves, we need to talk about their assumptions. In statistics, if you do not satisfy the assumptions for a particular procedure, you might not be able to trust the results.

In chapter 1, we talked about one assumption. We saw how inferential statistics and hypothesis tests require unbiased, representative samples. If you use a sampling methodology that produces a non-representative sample, you can't trust your hypothesis test results. The underlying data just don't represent the population you're studying. However, there are other assumptions you need to consider. All hypothesis tests have some assumptions. Many are similar between tests, but some are unique to particular methods.

Before getting to the tests, I need to make an essential point about graphing your data. As I discuss in my *Introduction to Statistics* book, data analysis is best when you use graphs and numeric analysis together. Charts bring your data to life in a way that statistical output cannot. Additionally, they are an easy way to identify potential problems, such as skewed distributions and outliers. While graphing is a crucial part of the process, I focus on the hypothesis tests themselves in this book.

1-Sample t-Tests

Use a one-sample t-test to compare your sample mean to a hypothesized value for the population and to create a confidence interval of likely values for the population mean. Typically, researchers use a hypothesized value that is meaningful for their study, which forms the null hypothesis for the test.

Please note that with a 1-sample t-test, you don't need to perform the hypothesis test. If you just want to understand the precision of the estimate, assess the confidence interval to identify the range of likely values. That information can be valuable even when you don't have a reference or hypothesized value.

In the energy cost example, researchers compared the current year's mean cost to last year's mean cost. The researchers used the sample data for the current year and entered the previous year's mean cost for the hypothesized value.

The 1-sample t-test has the following hypotheses:

- **Null**: The population mean equals the hypothesized mean.
- **Alternative**: The population mean does not equal the hypothesized mean.

If the p-value is less than your significance level (e.g., 0.05), you can reject the null hypothesis. The difference between the sample mean and the hypothesized mean is statistically significant. Your sample provides strong enough evidence to conclude that the population mean does not equal the hypothesized mean.

Assumptions

The assumptions for the 1-sample t-tests cover a variety of issues that range from data collection, data type, and properties of the data. For reliable 1-sample t-test results, your data should satisfy the following assumptions:

You have a random sample

Drawing a random sample from the population you are studying helps ensure that your data represent the population. Representative samples are vital when you want to make inferences about the population. If your data do not represent the population, your analysis results will not be valid for that population.

You must draw a random sample from your population of interest. Each item or person in the population must have an equal probability of being selected.

For example, if you're studying demographic information in a city and randomly approach people in a mall, the people probably do not accurately represent the entire state. People in that specific mall are likely to live within a particular geographic region and have distinct socioeconomic characteristics that don't represent the city's population.

Your data must be continuous

T-tests require continuous data. Continuous variables can take on any numeric value, and the scale can be meaningfully divided into smaller increments, including fractional and decimal values. There are an infinite number of possible values between any two values. And the difference between any two values is always meaningful. Typically, you measure continuous variables on a scale. For example, when you measure height, weight, and temperature, you have continuous data.

Other hypothesis tests can analyze different types of data. I'll cover those in later chapters.

Your sample data should follow a normal distribution or have more than 20 observations

T-tests assume that your data follow the normal distribution, which many natural phenomena follow. The normal distribution is that familiar symmetric, bell-shaped curve in statistics. However, when your sample is larger than 20, your data can be skewed and the test results will still be reliable. When your sample size is less than 20, graph your data and determine whether the distribution is skewed or has outliers. Either of these conditions can cause the test results to be unreliable.

Fortunately, if you have more than 20 observations, you don't have to worry about the normality assumption in most cases. In chapter 7, I discuss how the central limit theorem allows you to waive the normality assumption when your sample size is sufficiently large.

1-sample t-test example

Imagine we've conducted a simple experiment using a random sample of participants. We want to determine whether the participants have mastered a skill more than a target value after attending an educational session. After the session, we assess the skill level of 15 participants using a validated assessment.

We want to determine whether the mean score of these participants is different than the reference value of 60. A significantly higher mean represents an improvement. Download the CSV data file: Assessment Scores.

Let's analyze the data!

One-Sample T: Score

```
Test of μ = 60 vs ≠ 60


Variable   N   Mean  StDev  SE Mean       95% CI          T      P
Score     15  64.16  11.35     2.93  (57.87, 70.45)  1.42  0.178
```

The output indicates that the mean score is 64.16, which is higher than the reference value. For now, the p-value is the most important statistic. We'll learn about t-values and degrees of freedom in later chapters.

If the p-value is less than your significance level, the difference between the mean and reference value is statistically significant.

Because our p-value (0.178) is greater than the standard significance level of 0.05, we fail to reject the null hypothesis. If the p-value is high, the null will fly! Our sample data do not provide enough evidence to support the claim that the population mean is different from 60. We cannot conclude that the educational session was useful.

Furthermore, the sample estimate of the mean (64.16) is based on 15 observations and is unlikely to equal the population mean. The confidence interval estimates that the actual population mean is likely between 57.87 and 70.45. The confidence interval includes the reference value of 60, which is why we cannot conclude that the population mean is different from that value.

Later in this chapter, I discuss in detail what it means when we fail to reject the null hypothesis and why statisticians don't accept the null. For now, just know that we don't have enough evidence to conclude that the population mean is different from 60. However, this result doesn't prove that the population mean equals 60.

2-Sample t-Tests

Use two-sample t-tests to compare the means of precisely two groups—no more and no less! The procedure also creates a confidence interval of the mean difference. Typically, you perform this test to determine whether two population means are different.

For example, do students who learn using Method A have a different mean score than those who learn using Method B?

2-sample t-tests have the following hypotheses:

- **Null hypothesis:** The means for the two populations are equal.
- **Alternative hypothesis:** The means for the two populations are not equal.

If the p-value is less than your significance level (e.g., 0.05), you can reject the null hypothesis. The difference between the two means is statistically significant. Your sample provides strong enough evidence to conclude that the two population means are not equal.

Assumptions

The assumptions for 2-sample t-tests are similar to those for the 1-sample version. I'll focus on the differences below.

For reliable 2-sample t-test results, your data should satisfy the following assumptions:

- You have a representative, random sample
- Your data must be continuous

However, there are several differences between the 1-sample and 2-sample t-tests.

Your sample data should follow a normal distribution or each group has more than 15 observations

All t-tests assume that your data follow the distribution. However, as you saw for 1-sample t-tests, you can waive this assumption if your sample size is large enough.

For the 2-sample t-test, when each group is larger than 15, your data can be skewed and the test results will still be reliable. When your sample size is less than 15 per group, graph your data and determine whether the two distributions are skewed or has outliers. Either of these conditions can cause the test results to be unreliable.

Fortunately, if you have more than 15 observations in each group, you don't have to worry about the normality assumption too much.

The groups are independent

Independent samples contain different sets of items in each sample. 2-sample t-tests compare measurements taken on two distinct groups. If you have the same people or items in both groups, you can use the paired t-test.

Groups can have equal or unequal variances but use the correct form of the test

Variance, and the closely related standard deviation, are measures of variability. Each group in your analysis has its own variance. The two-sample t-test has two methods. One method assumes that the two groups have equal variances while the other does not assume they are equal. The form that does not assume equal variances is known as Welch's t-test.

When the sample sizes for both groups are equal, or nearly equal, and you have a moderate sample size, t-tests are robust to differences between variances. If one group has twice the variance of another group, it's time to use Welch's t-test! However, you don't need to worry about smaller differences.

If you have unequal variances *and* unequal sample sizes, it's vital to use the unequal variances version of the 2-sample t-test!

2-sample t-test example

Let's conduct a two-sample t-test! Our hypothetical scenario is that we are comparing scores from two teaching methods. We drew two random samples of students. Students in one group learned using Method A while the other group used Method B. These samples contain entirely separate students.

Now, we want to determine whether the two means are different. Download the CSV file that contains all data for the 2-sample t-test and the pair t-test examples: t-TestExamples.

Here is what the data look like in the datasheet.

Method A	Method B
72.471714	72.145335
72.100548	89.811362
69.700219	98.071997
61.294691	84.486978
76.509736	80.530738
81.288642	84.8586
75.898287	70.828394
71.610303	90.863795
82.005608	73.113933
52.743031	93.715501
64.720142	83.399928
89.807999	86.025116
74.101	92.191169
60.052173	91.876728
68.25018	79.216534

Let's assume that the variances are equal and use the Assuming Equal Variances version.

Two-Sample T-Test and CI: Method A, Method B

```
Two-sample T for Method A vs Method B

            N   Mean  StDev  SE Mean
Method A   15  71.50   9.41      2.4
Method B   15  84.74   8.31      2.1


Difference = μ (Method A) - μ (Method B)
Estimate for difference:  -13.24
95% CI for difference:  (-19.89, -6.59)
T-Test of difference = 0 (vs ≠): T-Value = -4.08  P-Value = 0.000  DF = 27
```

The output indicates that the mean for Method A is 71.50 and for Method B it is 84.74. Looking in the Standard Deviation column, we can see that they are not exactly equal, but they are close enough to assume equal variances.

Because our p-value (0.000) is less than the standard significance level of 0.05, we can reject the null hypothesis. If the p-value is low, the null must go! Our sample data support the claim that the population means are different. Specifically, Method B's mean is greater than Method A's mean. If high scores are better, then Method B is significantly better than Method A.

The sample estimate of the mean difference is -13.24. However, that estimate is based on 30 observations split between the two groups and it is unlikely to equal the population difference. The confidence interval estimates that the mean difference between these two methods for the entire population is likely between -19.89 and -6.59.

The negative values reflect the fact that Method A has a lower mean than Method B (i.e., Method A – Method B < 0). The confidence interval excludes the value of zero (no difference between groups), so we can conclude that the population rates are different.

Paired t-Tests

Use paired t-tests to assess dependent samples, which are two measurements on the same person or item.

Suppose you gather a random sample of people. You give them all a pretest, administer a treatment, and then perform a posttest. Each subject has a pretest and posttest score. Or, perhaps you have a sample of wood boards, and you paint half of each board with one paint and the other half with different paint. Then, you measure the paint durability for both types of paint on all the boards. Each board has two paint durability scores.

In both cases, you can use a paired t-test to determine whether the difference between the means of the two sets of scores is statistically significant.

Assumptions

For reliable paired t-test results, your data should satisfy the following assumptions:

- You have a representative, random sample
- Your sample contains independent observations
- Your data must be continuous
- Data should follow a normal distribution or have a sample size larger than 20.

Dependent Samples

Unlike 2-sample t-tests, paired t-tests use the same people or items in both groups. One way to determine whether a paired t-test is appropriate for your data is if each row in the dataset corresponds to one person or item.

Paired t-Test example

For this example, imagine that we have a training program. We need to determine whether the difference between the mean pretest score and the mean post-test score is significantly different.

Here is what the data look like in the datasheet. Note that the analysis does not use the subject's ID number.

SubjectID	Pretest	Posttest
1	90.563	110.642
2	94.816	101.588
3	109.56	120.607
4	90.222	83.2217
5	97.598	109.272
6	91.167	115.806
7	96.65	99.8958
8	97.616	117.94
9	88.845	106.052
10	90.817	82.8229
11	89.294	116.639
12	115.83	128.61
13	121.29	119.665
14	87.872	108.383
15	93.793	96.3738

Paired T-Test and CI: Pretest, Posttest

```
Paired T for Pretest - Posttest

              N     Mean  StDev  SE Mean
Pretest      15    97.06  10.31     2.66
Posttest     15   107.83  13.25     3.42
Difference   15   -10.77  11.17     2.88


95% CI for mean difference: (-16.96, -4.59)
T-Test of mean difference = 0 (vs ≠ 0): T-Value = -3.73  P-Value = 0.002
```

The output indicates that the mean for the Pretest is 97.06 and for the Posttest it is 107.83. The difference between the pretest and posttest is -10.77. If the p-value is less than your significance level, the difference does not equal zero.

Because our p-value (0.002) is less than the standard significance level of 0.05, we can reject the null hypothesis. Our sample data support the hypothesis that the population means are different. Specifically, the Posttest mean is greater than the Pretest mean.

Again, the sample estimate of the difference (-10.77) is unlikely to equal the population difference. The confidence interval estimates that the actual population difference between the Pretest and Posttest is likely between -16.96 and -4.59.

The negative values reflect the fact that the Pretest has a lower mean than the Posttest (i.e., Pretest – Posttest < 0). The confidence interval excludes the value of zero (no difference between groups), so we can conclude that the population rates are different.

If high scores are better, then the Posttest scores are significantly better than the pretest scores.

Paired t-Tests Are Really 1-Sample t-Tests

I've seen a lot of confusion over how a paired t-test works and when you should use it. Pssst! Here's a secret! Paired t-tests and 1-sample t-tests are the same hypothesis test incognito!

You use a 1-sample t-test to assess the difference between a sample mean and the null hypothesis value.

A paired t-test takes paired observations (like before and after), subtracts one from the other, and conducts a 1-sample t-test on the differences. Typically, a paired t-test determines whether the paired differences are significantly different from zero.

Download the CSV data file to check this yourself: T-testData. All of the statistical results are the same when you perform a paired t-test using the Before and After columns versus performing a 1-sample t-test on the Differences column.

↓	C1	C2	C3
	Before	After	Difference
1	-1.43233	1.00369	-2.43602
2	-1.43952	1.62507	-3.06459
3	-0.87813	-0.07925	-0.79888
4	2.04618	0.47962	1.56656
5	-0.06234	1.20517	-1.26751
6	0.83785	1.33710	-0.49925
7	0.60482	2.55502	-1.95020
8	1.64802	1.22170	0.42632

Paired T-Test and CI: Before, After

```
Paired T for Before - After

             N    Mean  StDev  SE Mean
Before      15   0.003  1.276    0.329
After       15   0.998  0.779    0.201
Difference  15  -0.995  1.575    0.407


95% CI for mean difference: (-1.867, -0.123)
T-Test of mean difference = 0 (vs ≠ 0): T-Value = -2.45  P-Value = 0.028
```

One-Sample T: Difference

```
Test of μ = 0 vs ≠ 0


Variable     N    Mean  StDev  SE Mean        95% CI             T      P
Difference  15  -0.995  1.575    0.407  (-1.867, -0.123)  -2.45  0.028
```

When you realize that paired t-tests are the same as 1-sample t-tests on paired differences, you can focus on the deciding characteristic — does it make sense to analyze the differences between two columns?

Suppose the Before and After columns contain test scores and there was an intervention in between. If each row in the dataset contains the same subject in the two columns, it makes sense to find the difference between the columns. It represents how much each subject changed after the intervention. The paired t-test is the correct choice.

On the other hand, if a row has different subjects in the two columns, it doesn't make sense to subtract the columns. You should use the 2-sample t-test.

The paired t-test is a convenience for you. It eliminates the need for you to calculate the difference between two columns yourself. Remember, double-check that this difference is meaningful!

Why Not Accept the Null Hypothesis?

In the one-sample t-test example earlier in this chapter, our p-value was greater than the significance level. Consequently, we failed to reject the null hypothesis.

Failing to reject the null hypothesis is an odd way to state that the results of your hypothesis test are not statistically significant. Why the peculiar phrasing? "Fail to reject" sounds like one of those double negatives that writing classes taught you to avoid. What does it mean exactly? There's an excellent reason for the odd wording!

In this section, learn what it means when you fail to reject the null hypothesis and why that's the correct wording. While accepting the null hypothesis sounds more straightforward, it is not statistically accurate!

Before proceeding, let's recap some necessary information. In all hypothesis tests, you have the following two hypotheses:

- The null hypothesis states that there is no effect or relationship between the variables.
- The alternative hypothesis states the effect or relationship exists.

We assume that the null hypothesis is correct until we have enough evidence to suggest otherwise.

After you perform a hypothesis test, there are only two possible outcomes.

- When your p-value is less than or equal to your significance level, you reject the null hypothesis. The data favors the alternative hypothesis. Congratulations! Your results are statistically significant.
- When your p-value is greater than your significance level, you fail to reject the null hypothesis. Your results are not significant. You'll learn more about interpreting this outcome later in this section.

To understand why we don't accept the null, consider that you can't prove a negative. A lack of evidence isn't proof that something doesn't exist. You just haven't proven that it exists. It might exist, but your study missed it. That's a huge difference, and it is the reason for the convoluted wording. Let's look at several analogies.

Species Presumed to be Extinct

Australian Tree Lobsters were assumed to be extinct. There was no evidence that any were still living because no one had seen them for decades. Yet in 1960, scientists observed them. The same thing happened to the Gracilidris Ant and the Nelson Shrew, among many others. Dedicated scientists were looking for these species but hadn't been in the right time and place to observe them. Lack of proof doesn't represent proof that something doesn't exist!

Criminal Trials

In a trial, we start with the assumption that the defendant is innocent until proven guilty. The prosecutor must work hard to exceed an evidentiary standard to obtain a guilty verdict. If the prosecutor does not meet that burden, it doesn't prove the defendant is innocent. Instead, there was insufficient evidence to conclude he is guilty.

Perhaps the prosecutor conducted a shoddy investigation and missed clues? Or, the defendant successfully covered his tracks? Consequently, the verdict in these cases is “not guilty.” That judgment doesn’t say the defendant is proven innocent, just that there wasn’t enough evidence to move the jury from the default assumption of innocence.

Hypothesis Tests

When you’re performing hypothesis tests in statistical studies, you typically want to find an effect or relationship between variables. The default position in a hypothesis test is that the null hypothesis is correct. Like a court case, the sample evidence must exceed the evidentiary standard, which is the significance level, to conclude that an effect exists.

The hypothesis test assesses the evidence in your sample. If your test fails to detect an effect, that’s not proof it doesn’t exist. It just means your sample contained an insufficient amount of evidence to conclude that it exists. Like the “extinct” species, or the prosecutor who missed clues, the effect might exist in the overall population but not in your particular sample. Consequently, the test results fail to reject the null hypothesis, which is analogous to a “not guilty” verdict in a trial. There just wasn’t enough evidence to move the hypothesis test from the default position that the null is true.

The critical point across these analogies is that a lack of evidence does not prove something does not exist—just that you didn’t find it in your specific investigation. Hence, you never accept the null hypothesis.

Interpreting Failures to Reject the Null

Accepting the null hypothesis would indicate that you’ve proven an effect doesn’t exist. As you’ve seen, that’s not the case at all. You can’t prove a negative! Instead, the strength of your evidence falls short of being able to reject the null. Consequently, we fail to reject it.

Failing to reject the null indicates that our sample did not provide sufficient evidence to conclude that the effect exists. However, at the same time, that lack of evidence doesn't prove that the effect does not exist. Capturing all that information leads to the convoluted wording! What are the possible interpretations of failing to reject the null hypothesis? Let's work through them.

First, it is possible that the effect indeed doesn't exist in the population, so your hypothesis test didn't detect it in the sample. Makes sense, right? While that *is* one possibility, it doesn't end there.

Another possibility is that the effect exists in the population, but the test didn't detect it for a variety of reasons. These reasons include the following:

- The sample size was too small to detect the effect.
- The variability in the data was too high. The effect exists, but the noise in your data swamped the signal (effect).
- By chance, you collected a fluky sample. When dealing with random samples, chance always plays a role in the results. The luck of the draw might have caused your sample not to reflect an effect that exists in the population.

Notice how studies that collect a small amount of data or low-quality data are likely to miss an effect that exists? These studies had an inadequate ability to detect the effect. We certainly don't want to take results from low-quality studies as proof that something doesn't exist!

However, failing to detect an effect does not necessarily mean a study is low-quality. Random chance in the sampling process can work against even the best research projects!

Using Confidence Intervals to Compare Means

We're shifting gears to close this chapter. The previous sections showed you different ways to compare means using t-tests. This section shows you one way NOT to compare two means that I've seen people use too often.

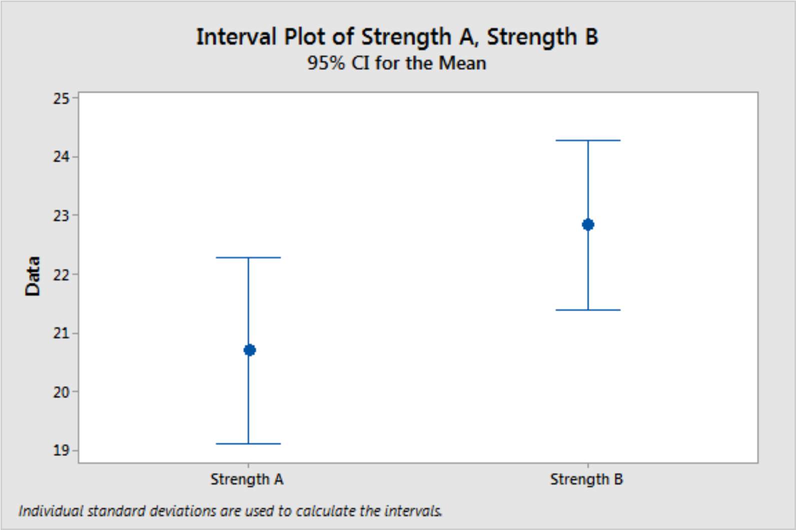
Analysts often compare the confidence intervals for two groups to determine whether the difference between two means is statistically significant. If those intervals overlap, they conclude that the difference between groups is not statistically significant. If there is no overlap, the difference is significant.

While this visual method of assessing the overlap is easy to perform, it regrettably reduces your ability to detect differences. Fortunately, there is a simple solution to this problem that allows you to perform a simple visual assessment and yet not diminish the power of your analysis.

I'll start by showing you the problem in action and explain why it happens. Then, we'll proceed to an easy alternative method that avoids this problem.

Determining whether confidence intervals overlap is an overly conservative approach for identifying significant differences between groups. It's true that when confidence intervals don't overlap, the difference between groups is statistically significant. However, when there is some overlap, the difference might still be significant.

Suppose you're comparing the mean strength of products from two groups and graph the 95% confidence intervals for the group means, as shown below. Download the CSV dataset that I use throughout this section: DifferenceMeans.



Jumping to Conclusions

Upon seeing how these intervals overlap, you conclude that the difference between the group means is not statistically significant. After all, if they're overlapping, they're not different, right?

This conclusion sounds logical, but it's not necessarily true. In fact, for these data, the 2-sample t-test results are statistically significant with a p-value of 0.044. Despite the overlapping confidence intervals, the difference between these two means is statistically significant.

This example shows how the CI overlapping method fails to reject the null hypothesis more frequently than the corresponding hypothesis test. Using this method decreases your ability to detect differences, potentially causing you to miss essential findings.

This apparent discrepancy between confidence intervals and hypothesis test results might surprise you. Analysts expect that confidence intervals with a confidence level of (100 – X) will always agree with a hypothesis test that uses a significance level of X percent. For example, analysts often pair 95% confidence intervals with tests that use a

5% significance level. It's true. Confidence intervals and hypothesis tests should always agree. So, what is happening in the example above?

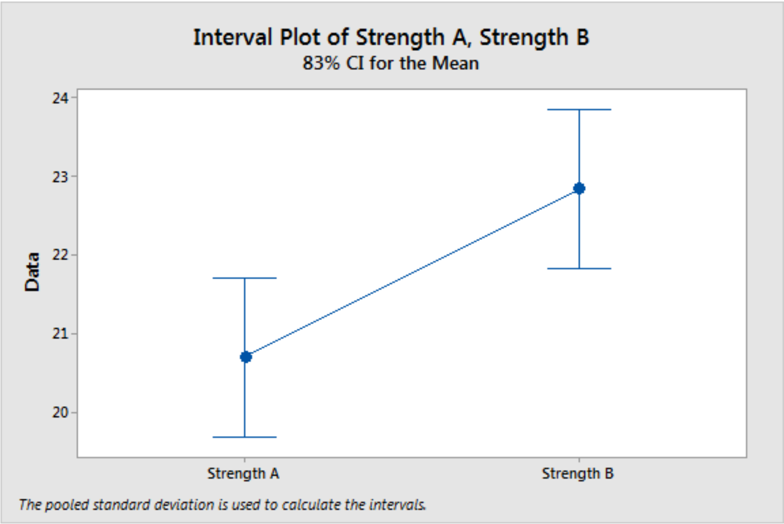
Using the Wrong CIs

The problem occurs because we are not comparing the correct confidence intervals to the hypothesis test result. The test results apply to the difference between the means while the CIs apply to the estimate of each group's mean—not the difference between the means. We're comparing apples to oranges, so it's not surprising that the results differ.

To obtain consistent results, we must use confidence intervals for differences between group means—we'll get to those CIs shortly.

However, if you doggedly want to use CIs of each group to make this determination, there are several possible methods.

Goldstein and Healy find that for barely non-overlapping intervals to represent a 95% significant difference between two means, use an 83% confidence interval of the mean for each group. The graph below uses this confidence level for the same dataset as above, and they don't overlap. (Goldstein & Healy, 1995)



Cumming and Finch find that the degree of overlap for two 95% confidence intervals for independent means allows you to estimate the p-value for a 2-sample t-test when sample sizes are greater than 10. When the confidence limit of each CI reaches approximately the midpoint between the point estimate and the limit of the other CI, the p-value is near 0.05. (Cumming & Finch, 2005)

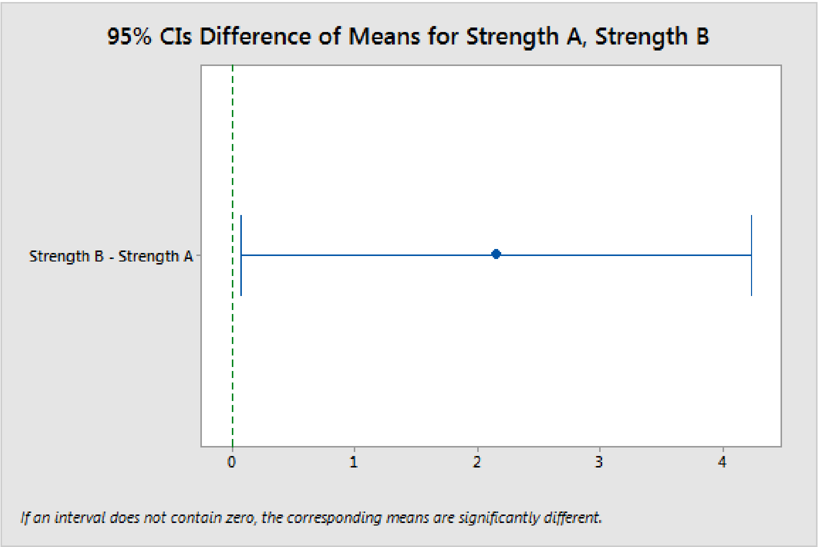
The first graph in this section, with the 95% CIs, approximates this condition, and the p-value is near 0.05. Lower amounts of overlap correspond to lower p-values. For 95% CIs, when the end of one CI just reaches the end of the other CI, it corresponds to a p-value of about 0.01.

To me, these approaches seem kludgy. Using a confidence interval of the difference is a more straightforward solution that even provides additional useful information.

Confidence Intervals of Differences

Previously, we saw how the apparent disagreement between the group CIs and the 2-sample test results occurs because we used the wrong confidence intervals. Instead, we need a CI for the difference between group means. This type of CI will always agree with the 2-sample t-test—just be sure to use the equivalent combination of confidence level and significance level (e.g., 95% and 5%). We're now comparing apples to apples!

Using the same dataset, the confidence interval below presents a range of values that likely contains the mean difference for the entire population. The interpretation continues to be a simple visual assessment. Zero represents no difference between the means. Does the interval contain zero? If it does not include zero, the difference is statistically significant because the range excludes no difference. At a glance, we can tell that the difference is statistically significant.



This graph corresponds with the 2-sample t-test results below. Both test the difference between the two means. This output also provides a numerical representation of the CI of the difference [0.06, 4.23].

Two-Sample T-Test and CI: Strength B, Strength A

```
Two-sample T for Strength B vs Strength A

             N   Mean  StDev  SE Mean
Strength B  20  22.84   3.08     0.69
Strength A  20  20.69   3.41     0.76


Difference = μ (Strength B) - μ (Strength A)
Estimate for difference:  2.15
95% CI for difference:  (0.06, 4.23)
T-Test of difference = 0 (vs ≠): T-Value = 2.09  P-Value = 0.044  DF = 37
```

In addition to providing a simple visual assessment, the confidence interval of the difference presents crucial information that neither the group CIs nor the p-value provides. It answers the question, based on our sample, how large is the difference between the two populations likely to be? Like any estimate, there is a margin of error around the point estimate of the difference. It's important to factor in this margin of error before acting on findings.

For our example, the point estimate of the mean difference is 2.15, and we can be 95% confident that the population difference falls within the range of 0.06 to 4.23.

As with all CIs, the width of the interval for the mean difference reveals the precision of the estimate. Narrower intervals suggest a more precise estimate. And, you can assess whether the full range of values is practically significant.

When the interval is too wide (imprecise) to be helpful and/or the range includes differences that are not practically significant, you have reason to hesitate before making decisions based on the results. These types of CI results indicate that you might not obtain meaningful benefits even though the difference is statistically significant.

There's no statistical method for answering questions about how precise an estimate must be or how large an effect must be to be practically useful. You'll need to apply your subject-area knowledge to the confidence interval of the difference to answer these questions.

For the example in this section, it's important to note that the low end of the CI is very close to zero. It will not be surprising if the actual population difference falls close to zero, which might not be practically significant despite the statistically significant result. If you are considering switching to Group B for a stronger product, the mean improvement might be too small to be meaningful.

When you're comparing groups, assess confidence intervals of those differences rather than comparing confidence intervals for each group. This method is simple, and it even provides you with additional valuable information.

Review and Next Steps

You're already interpreting test results from three different types of t-tests! Not too bad!

We also looked at the assumptions you need to consider for all these tests. If your data collection methodology and the properties of your data don't satisfy these assumptions, you might not be able to trust your results. However, it's important to remember that some assumptions are strong requirements (i.e., representative sample), while you can waive others when your sample is large enough (i.e., normality assumption).

In this chapter, we mainly looked at the sample means, p-values, and confidence intervals. However, I'm sure you noticed there are other values in the output. While we looked at the most important statistics for determining statistical significance, the other statistics represent crucial aspects of how the tests work. The upcoming chapters will systematically explain these additional statistics.

You learned that we do not accept the null hypothesis. Instead, we fail to reject it. The convoluted wording encapsulates the fact that insufficient evidence for an effect in our sample isn't proof that the effect does not exist in the population. The effect might exist, but our sample didn't detect it—just like all those species scientists presumed were extinct because they didn't see them.

Finally, I admonished you not to use the common practice of seeing whether confidence intervals of the mean for two groups overlap to determine whether the means are different. That process causes you to lose statistical power and miss out on important information about the likely range of the mean difference. Instead, assess the confidence interval of the difference between means.

One of the statistics in the output for t-tests is the t-value, which is a test statistic. T-tests are named after the t-values and t-distributions they use to determine statistical significance. In chapter 3, you learn about test statistics, sample sizes, data variability, and how they all relate to sampling distributions and the outcome of the hypothesis test.

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